

TeleClinic Platform — Month 3 Learning Review

Candidate: Twahirwa, Herve

Role: Data Analyst & MEL Associate · **Period:** Feb–Apr 2026 (Month 3)

Code repository: [INSERT YOUR PUBLIC GITHUB REPO URL]

Part 1: Data Quality Audit

I started with the data dictionary, then ran missingness checks, duplicate primary-key tests, and join checks across all seven tables before calculating any performance numbers. Below is every material data quality issue I found, with what I think caused it and what it means for reporting.

#	Issue	Likely explanation	Impact on reporting
1	18 referrals (17.6%) with missing authorisation status	Referral initiated clinically but authorisation never recorded or did not sync (NULL = missing record)	RM-01/02 metrics unreliable; processing time cannot be calculated
2	122 completed consults without notes; 169 without ICD	Clinicians closing under time pressure; no hard stop before sign-off	Clinical audit weakened; quality metrics must use completed consults only
3	307 rows with null diagnosis/notes/ICD/call_type (27.6%)	Expected for cancelled/no-show bookings — not a data entry bug	Misleading if included in clinical quality denominators
4	213 incomplete registrations / failed validation	Insurance API: NID not found, expired cover, timeouts	Onboarding funnel incomplete; 491 registered patients never booked
5	62 lab results uploaded but not viewed by clinician	No alert when results sit unreviewed	Patient safety risk — results may never reach patient
6	23 lab tests without uploaded results	Lab capacity, sample issues, or sync delay	Lab completion and TAT metrics incomplete
7	81 prescriptions not dispensed	Non-collection, stock-out, or care pathway change	Care continuity gap; hard to separate clinical vs operational reasons
8	15 referrals not authorised + 18 unknown status	Triage rejection or admin backlog at receiving facility	Patient may not know referral was denied or stalled
9	failure_reason null on 987 insurance log rows	Field only populated on failed attempts	Fine if filtered to failures; misleading if reported raw

Join problem

I checked every foreign key link (patient_id, consultation_id, original_consult_id). All IDs match cleanly — zero orphan records. The deliberate referral issue is different: 18 referrals have a valid consultation_id but NULL authorised, authorisation_datetime, and processing_hours. I treated these as incomplete workflow records, not bad joins. For referral authorisation rate I used only records with known status (69 authorised out of 84 known = 82.1%), and reported the 18 unknowns separately rather than dropping them.

Low-volume district: Musanze

Musanze stands out: only 6 consultations against 21 registrations (coverage ratio 0.29), the lowest of all 12 districts. Three plausible explanations: (1) later rollout in the 13-week window; (2) rural connectivity and awareness — all 21 Musanze patients are rural; (3) small-number volatility with only 21 registrations. To investigate I would need district activation dates, clinician roster hours, outreach logs, and network coverage data.

Top patient safety concern

The 62 uploaded lab results that clinicians never viewed. In telemedicine the platform is the only bridge between lab and patient — a patient who tests and hears nothing may assume all is well.

Part 2: Metrics & Analysis

I picked five metrics that together cover access, clinical quality, patient safety, care continuity, and the onboarding barrier — without duplicating the same story twice.

Metric	Result	Definition & exclusions	Why I chose it
Consultation completion rate	72.4% (807/1,114)	Completed ÷ all booked consults	Shows whether demand converts to care; 14.9% no-show is lost capacity
Full clinical documentation rate	79.1% (638/807)	Notes + ICD both Yes; completed consults only	Care that cannot be audited is hard to govern
Lab result clinician review rate	82.3% (288/350)	Clinician viewed = Yes; uploaded results only	Unreviewed results are a direct safety gap
Prescription dispensing rate	84.5% (440/521)	Dispensed Yes ÷ all prescriptions	Remote prescribing only helps if pharmacy dispenses
First-attempt insurance validation	82.2%	Success on attempt_number = 1	Front-door barrier; Uninsured succeeds only 9.9%

What this data cannot tell us

Clinical outcomes, patient satisfaction, whether referred patients reached facilities, clinical appropriateness of diagnoses/Rx, or whether notes are clinically adequate.

Result that concerns me most

Roughly one in five care episodes has incomplete documentation or an unreviewed lab result. The 62 unreviewed results are the clearest danger — telemedicine has no fallback if the clinician does not open the result. The platform is being used (1,114 bookings) but has not closed the loop on results management.

Part 3: Equity Analysis

Equity measures calculated

1. Rural Representation Index (RRI): rural share ÷ 83% national rural share. Registrations and completed consults: 60.1% rural → RRI = 0.72.

2. District coverage ratio: consultations ÷ registrations by district. Musanze 0.29 to Rwamagana 1.09.

Gender near-balanced (50.2% male registrants). USSD ~41% of rural consults. Insurance sharpest gap: Uninsured validation 9.9% vs 78–90% formal schemes.

Gates Foundation quarterly report paragraph

After three months, TeleClinic registered 1,200 patients and completed 807 consultations across 12 districts. Rural users account for 60% of activity vs ~83% nationally (RRI = 0.72). USSD carries ~41% of rural consults — encouraging for feature-phone users — but urban districts especially Gasabo dominate volume. Gender balanced. The equity gap I would flag: uninsured patients succeed on validation less than 10% of the time. District coverage uneven (Musanze 0.29). **Recommended action:** USSD outreach in five lowest-coverage districts plus 72-hour provisional access for Uninsured patients; track RRI monthly toward 0.85 by Month 6.

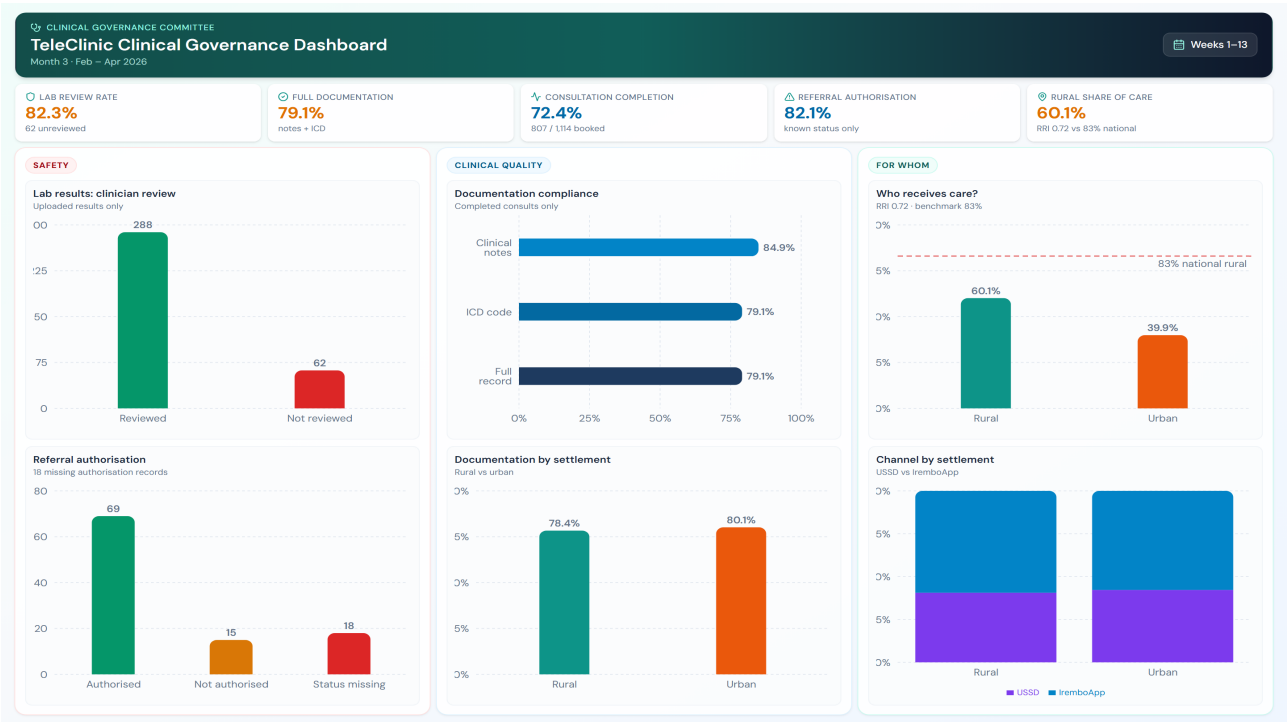
Equity dimension we cannot measure

Socioeconomic vulnerability — insurance scheme but not income or ability to pay. Add optional household quintile at registration and track validation failure → consult conversion.

Part 4: Dashboard Design

Audience: Clinical Governance Committee · **Central question:** Is the platform delivering safe, quality care — and for whom?

One-page React dashboard (teleclinic-dashboard/ in repo). Layout: KPI strip + Safety | Clinical Quality | For Whom. Screenshot below.



Element	Why this chart	Decision it supports
KPI strip	Five numbers give quick safe/quality/reach read	Whether Month 3 needs intervention now
Lab clinician review	62 unreviewed shown as count not just rate	When to escalate to clinical supervisor
Referral authorisation	Authorised / Not authorised / Missing (18)	Fix workflow before scaling referrals

Element	Why this chart	Decision it supports
Documentation compliance	Notes, ICD, full record side-by-side	Require ICD before consult close?
Documentation by settlement	Quality linked to rural vs urban	Do gaps cluster geographically?
Settlement share + 83% line	Rural under-representation visible	Shift roster or outreach to rural districts
Channel by settlement	USSD vs IremboApp in rural and urban care	Invest in USSD-first workflows?

Metric excluded: First-attempt insurance validation (82.2%) — onboarding metric for Gates/MOH ops, not Clinical Governance.

Part 5: Learning Review Brief

For the Gates Foundation quarterly call

Platform Performance at Month 3

- **People are using it.** 1,200 registrations, 1,114 bookings, 807 completed (72.4%).
- **Pathways partly close, safety gaps remain.** 84.5% Rx dispensed; 62 lab results never reviewed.
- **Rural reach below parity.** 60.1% rural vs ~83% national; Musanze lags.

What We Cannot Yet Conclude

Whether telemedicine improves outcomes; whether referrals reach facilities; whether poorest benefit most; whether volume is sustainable — 13 weeks too short.

One Recommended Change

Automated pathway closure tracker: flag unreviewed labs after 24h, undispensed Rx after 48h, unauthorised referrals after 72h — escalate to clinical supervisor.

Priority Question for Months 4–6

Does closing the lab and referral loop reduce care discontinuation among rural USSD users?

Reflection: North Star Metric

Documented Care Completion Rate (DCCR): share of booked consultations reaching Completed with notes and ICD both recorded. Month 3: **638 / 1,114 = 57.3%**.

Completion alone (72.4%) overstates success; documentation among completed (79.1%) hides bookings that never became care. DCCR combines both. Unlike lab or dispensing rates it applies to every consult; unlike RRI it measures delivery quality not reach.

Blind spots: no check on clinical correctness, unreviewed labs (62), undispensed Rx (81), equity-blind. Still, 57.3% honestly describes Month 3: live and used, not yet reliably delivering governable care at scale.

Process Note

I prioritised the data quality audit before performance numbers — unreviewed labs and the referral gap shaped the whole review. I spent less time on follow-ups; skipped forecasting. Most uncertain about Musanze. Built notebook + React dashboard for Part 4 governance view. With another day I'd add a pathway-closure metric chaining consult → lab → review → Rx.